4COSC002W Mathematics for Computing

Lecture 4

Elementary Logics

UNIVERSITY OF WESTMINSTER

What is Logic?

Logic is a system or principles guiding our thinking and reasoning. It enables us to classify information as *true* or *false* based on established rules and principles.

Elementary logic is not just about deducing whether a statement is true or false. It is the backbone of many areas of mathematics, philosophy, computer science, and even our day-to-day reasoning.

The origins of logic date back to ancient civilisations, where philosophers like Aristotle sought to understand the principles of valid reasoning.

Importance of Logic

In Mathematics

- Foundation for Mathematical Proofs
- Development of Abstract Thinking
- Ensuring Consistency (verifies for contradictions)

Real-World Applications of Logic

- **Computer Science:** Computer programs use logic gates and Boolean algebra, which are directly derived from elementary logic principles, to function and make decisions.
- **Philosophy:** Philosophers use logic to formulate arguments, understand concepts, and dissect philosophical problems.
- **Daily Decision-Making:** Every day, we use logic to make decisions, solve problems, and reason through various situations. For instance, if it looks cloudy outside, you might deduce that it will rain and decide to carry an umbrella.

Basic Concepts

Boolean values are the simplest form of logic: True (T) and False (F). These values are used to evaluate statements and operations in logic.

Logical Operators:

- **AND (∧):** Both statements must be true for the result to be true.
- **OR (∨):** At least one statement must be true for the result to be true.
- **NOT (¬):** Inverts the truth value of the statement.
- **IMPLIES (→):** If the first statement is true, the second must also be true for the result to be true.

A proposition (p) is a statement that is either true or false.

Propositions

We will consider two types of prepositions (statements):

- **Primitive or atomic** (consist of one preposition)
- **Compound** (consists of one preposition and logical operator, or two or more prepositions)

For example:

"$2 \times 2 = 5$" is a primitive statement which is *false*.

Proposition Examples

Primitive Propositions (Statements)

Abbreviation	**Proposition**	**Truth Value**
P1	Python is a programming language.	true
P2	Google Chrome is an antivirus software.	false
P3	USB stands for Universal Serial Bus.	true

Compound Propositions (Statements)

Abbreviation	**Proposition**	**Components**	**Truth Value**
C1	Python is a programming language **AND** USB stands for Universal Serial Bus.	P1 AND P3	true
C2	Google Chrome is an antivirus software **OR** Python is a programming language.	P2 OR P1	true
C3	**NOT** (Google Chrome is an antivirus software).	NOT P2	true

Truth Tables

Truth tables are tabular representations that list all possible truth values of logical expressions.

They are helpful in understanding and visualising how different *logical operators* work.

We will consider the **four basic truth tables.**

Truth Table: Negation (¬, NOT)

Atomic proposition	Resulting Value
A	**¬A**
T	F
F	T

Truth Table: Conjunction (∧, AND)

Atomic propositions		Resulting Value
A	**B**	**A∧B**
T	T	T
T	F	F
F	T	F
F	F	F

Truth Table: Disjunction (∨, OR)

Atomic propositions		Resulting Value
A	B	A∨B
T	T	T
T	F	T
F	T	T
F	F	F

Truth Table: Implication (→; IF…, THEN)

Atomic propositions		Resulting Value
A	**B**	**A→B**
T	T	T
T	F	F
F	T	T
F	F	T

Order of Precedence in Logical Operations

1. **Negation** (¬, NOT)
2. **Conjunction** (∧, AND)
3. **Disjunction** (∨, OR)
4. **Implication** (→, IMPLIES)

When an expression contains operations with the same precedence level, they are typically evaluated *from left to right*.

However, *parentheses* can be used to change the order of evaluation if necessary. Operations inside parentheses are always evaluated before those outside.

For example, in the expression **p∨q∧r**
the conjunction would be evaluated before the disjunction because of its higher precedence.
If you wanted to evaluate the disjunction before the conjunction, you could use parentheses: **(p∨q)∧r**

Algorithm of Computing Truth Tables

Let a formula C consists of the atomic propositions $p_1, p_2, \ldots, p_n$.

To compute the truth table, do the following:

1. List its *atomic propositions* $p_1, p_2, \ldots, p_n$ and count their number, n.
2. Form a table with $\boldsymbol{m = 2^n}$ rows
3. List all possible *combinations of the truth values* for $p_1, p_2, \ldots, p_n$ (following our new algorithms based on alternating periods of T and F)
4. *Prioritise the logical operations* in C, finding the main logic operation, etc.
5. Compute these operations in order.

Example 1: $(p \wedge q) \rightarrow p$

- The number of atomic statements: $n=2$
- The number of possible combinations of truth values: $2^n=2^2=4$
- The number of operations: 2, conjunction, then implication

Example 1: $(p \wedge q) \rightarrow p$

p	q		

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p	q		
T	T		
T	F		
F	T		
F	F		

Example 1: $(p \wedge q) \rightarrow p$

p	q	(1) $p \wedge q$ (conjunction)	(2) $(p \wedge q) \rightarrow p$ (implication)
T	T		
T	F		
F	T		
F	F		

Example 1: $(p \wedge q) \to p$

p	q	(1) p∧q (conjunction)	(2)
T	T	T	
T	F	F	
F	T	F	
F	F	F	

A	B	A∧B
T	T	T
T	F	F
F	T	F
F	F	F

Example 1: $(p \wedge q) \rightarrow p$

p	q	(1) $p \wedge q$	(2) $(p \wedge q) \rightarrow p$ (implication)
T	T	T	**T**
T	F	F	**T**
F	T	F	**T**
F	F	F	**T**

A	B	A→B
T	T	T
T	F	F
F	T	T
F	F	T

Example 2: $p \wedge \neg p$

p	(1) $\neg p$	(2) $p \wedge \neg p$
T	F	F
F	T	F

Example 3: $(p \wedge q) \rightarrow (p \vee r)$

p	q	r	(1) $p \wedge q$	(2) $p \vee r$	(3) $(p \wedge q) \rightarrow (p \vee r)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	F	T	T
F	F	F	F	F	T

Classification of Formulae

- In Logic, a formula *A* is called *SATISFIABLE* if there exists at least one output value T (true) in the truth table for *A*.
- In Logic, a formula *A* is called *UNSATISFIABLE* if there are no output values T (true) in the truth table for *A*.
- In Logic, a formula is called *VALID* if its output values in the truth table are all T (true).

<table>
<tr><td>SATISFIABLE AND VALID</td><td rowspan="2">UNSATISFIABLE</td></tr>
<tr><td>SATISFIABLE AND NOT VALID</td></tr>
</table>

Reasoning and Logical Consequence

Definition:

B is *a logical consequence* of *a knowledge base* A_1, A_2, A_3, ... A_n if the following formula is valid:

$$(A_1 \wedge (A_2 \wedge (A_3 \wedge \ldots A_n))) \Rightarrow B$$

Knowledge base　　　　Logical Consequence

Informally: we form an *implicative statement* with all members of the knowledge base joint by conjunction on the left-hand side and B on the right-hand side.

ACTIVITY: Good Tenants Problem

A landlady has a room to let

Two conditions: tenants should not drink and should not smoke.
Two students came.

- **A** came and said: *If I drink, then I smoke, but I do not drink*
- **B** came and said: *If I drink, then I smoke, but I do not smoke*

Who is the one to get this room?

Good Tenants Problem: Problem Analysis

Knowledge Base

Primitives:

- I drink $\boldsymbol{p}$
- I smoke $\boldsymbol{q}$

A came and said: *If I drink, then I smoke, but I do not drink*

$$(\boldsymbol{p} \Rightarrow \boldsymbol{q}) \wedge \neg \boldsymbol{p}$$

B came and said: *If I drink, then I smoke, but I do not smoke*

$$(\boldsymbol{p} \Rightarrow \boldsymbol{q}) \wedge \neg \boldsymbol{q}$$

Good Tenants Problem: Problem Analysis

(1) A came and said: If I drink, then I smoke, but I do not drink

$$(p \Rightarrow q), \neg p$$

What we want to check is if A *also does not smoke, i.e. we need to establish if from* $(p \Rightarrow q), \neg p$ we can conclude $\neg q$

(2) B came and said: If I drink, then I smoke, but I do not smoke

$$(p \Rightarrow q), \neg q$$

What we want to check is if ***B*** *also does not smoke, i.e we need to establish if from* $(p \Rightarrow q), \neg q$ we can conclude $\neg p$

Good Tenants Problem: Problem Analysis

What we want to check is if A *also does not smoke, i.e. we need to establish if from* $(p \Rightarrow q), \neg p$ we can conclude $\neg q$

This is a problem of logical consequence of $\neg q$

From the knowledge base $(p \Rightarrow q), \neg p$

Solution: form a formula

$$\boldsymbol{((p \Rightarrow q) \wedge \neg p) \Rightarrow \neg q}$$

Build and compute a truth table

Truth table for $((p \Rightarrow q) \land \neg p) \Rightarrow \neg q$

p	q	¬p	p⇒q	(p⇒q)∧¬p	¬q	((p⇒q)∧¬p)⇒¬q
T	T	F	T	F	F	T
T	F	F	F	F	T	T
F	T	T	T	T	F	F
F	F	T	T	T	T	T

NOT VALID, so there is no logical consequence between the knowledge base
$(p \Rightarrow q)$, $\neg p$ and $\neg q$

Good Tenants Problem: Problem Analysis

What we want to check is if B *also does not smoke, i.e we need to establish if from* $(p \Rightarrow q), \neg q$ we can conclude $\neg p$

This is a problem of logical consequence of $\neg p$

From the knowledge base $(p \Rightarrow q), \neg q$

Solution: form a formula

$$\boldsymbol{((p \Rightarrow q) \land \neg q) \Rightarrow \neg p}$$

Build and compute a truth table

Truth table for $((p \Rightarrow q) \wedge \neg p) \Rightarrow \neg p$

p	q	p⇒q	¬q	(p⇒q)∧¬q	¬p	((p⇒q)∧¬q)⇒¬p
T	T	T	F	F	F	**T**
T	F	F	T	F	F	**T**
F	T	T	F	F	T	**T**
F	F	T	T	T	T	**T**

VALID! So, there is a logical consequence between the knowledge base $(p \Rightarrow q)$, $\neg q$ and $\neg p$